

MF810 ADVANCED PROGRAMMING: DATA STRUCTURES AND ALGORITHMS

Improving Credit Default Prediction via NLP-Based Sentiment Analysis

A Hybrid Framework Integrating Structured & Unstructured Data

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Problem & Research Goals

The Problem

Traditional credit scoring (Basel II/III) relies heavily on structured financial variables like Debt-to-Income ratios and credit scores.

This approach ignores valuable "soft information" embedded in unstructured borrower text, such as loan descriptions and job titles.

Research Questions

Goal 1: Build a complete Expected Loss (EL) framework (PD, LGD, EAD).

Goal 2: Evaluate if NLP-derived features (Sentiment, TF-IDF) significantly improve predictive performance (AUC, Gini) over the baseline.

Expected Loss (EL)

IN CREDIT RISK MANAGEMENT

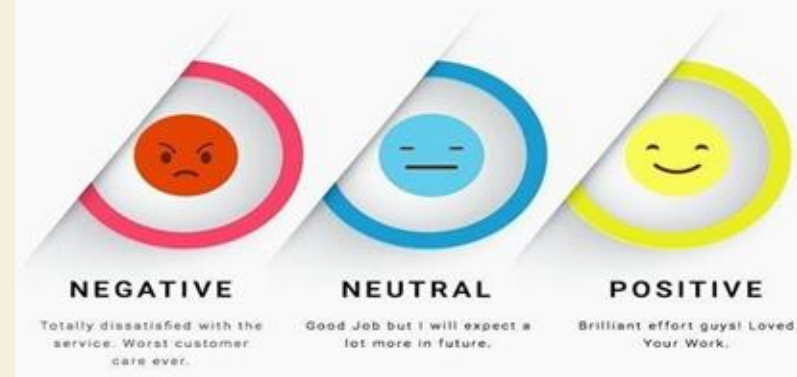
$$EL = PD \times EAD \times LGD$$

PD = Probability of Default

EAD = Exposure at Default

LGD = Loss Given Default

SENTIMENT ANALYSIS



Data Description

Lending Club Loan Dataset (2007–2018)



Dataset Overview

- ~**2260701** total observations
- ~**151** variables per record
- Includes both **structured** and **unstructured** data types

```
RangeIndex: 2260701 entries, 0 to 2260700  
Columns: 151 entries, id to settlement_term  
dtypes: float64(113), object(38)  
memory usage: 2.5+ GB
```



Key Variables

TARGET VARIABLE

• `loan_status`

→ Converted to Binary (Default vs. Non-Default)

STRUCTURED FEATURES

• `loan_amnt` `int_rate`

• `annual_inc` `dti`

• `revol_util`

UNSTRUCTURED FEATURES

• `desc` (Borrower loan description)

• `emp_title`

• `title`

Methodology: Baseline vs. Hybrid Models

Infrastructure: Dockerized Jupyter Environment | PySpark Processing | Parquet Storage



Baseline Model

Structured Only

FEATURE ENGINEERING

- Missing Value Imputation
- Categorical Encoding via **Weight of Evidence (WoE)**
- Feature Selection via **Information Value (IV)**

PREDICTIVE ALGORITHMS

- **PD:** Logistic Regression
- **LGD:** Beta Regression (Bounded 0-1)
- **EAD:** Linear Regression (CCF)



Hybrid Model

Structured + NLP

ADVANCED NLP PREPROCESSING

- + Text Cleaning (HTML removal, Stopwords, Lowercasing)
- + **Occupational Clustering** for high-cardinality titles

NLP FEATURE EXTRACTION

- + **Sentiment Analysis:** Psychological proxy scores $[-1, 1]$
- + **TF-IDF Vectorization:** Captures topical risk signals (e.g., "debt", "medical")
- + **Integration:** NLP features fused into Logistic Regression PD model



AUC (ROC)



KS Statistic



Lift Analysis



Gini Coefficient

$$EL = PD \times LGD \times EAD$$

Data Complexity & Challenges

Data Sparsity in Textual Features:

The primary textual field (**desc**) suffered from a 94% missing rate, rendering standard NLP feature extraction techniques ineffective due to the lack of usable data

High Cardinality in Categorical Data:

The employment title feature (**emp_title**) contained over 512,000 unique values, creating a high-cardinality problem that risked dimensionality issues and overfitting without proper standardization.

Entity Resolution Complexity:

The dataset included 120,000 joint applications, making it difficult to accurately model risk profiles at the aggregated household level using standard individual-level assessment methods.

Non-Linearity in Loss Distribution:

The Loss Given Default (LGD) exhibited complex, non-linear behavior (likely zero-inflated or bimodal), which traditional linear regression models failed to capture accurately.

